Referee Verification and Editorial Flags

On the 3D contour-generator Willmore-type estimate

Source note attributed to İmamverdi Bədəlov. This audit is not a certification of novelty or a machine-checked proof.

Decision

Main mathematical statement: VERIFIED.
Two-page proof as originally submitted: FLAG — incomplete; repaired in the accompanying paper.
Sharpness and equality: VERIFIED for the stated smooth, compact, boundaryless setting.
Projected-length identity: VERIFIED as a mathematical derivation; original attribution and novelty unverified.
Novelty / journal submission: QUALIFIED — priority research and human authorship review needed.

The audited upper bound is

$$\mathcal{L}(\Sigma)^2 \le \frac{A(\Sigma)}{2}\big(\mathcal{W}(\Sigma) - \pi\chi(\Sigma)\big), \quad \mathcal{W} = \int_\Sigma H^2\, dA, \quad H = \frac{\kappa_1 + \kappa_2}{2}.$$

The coefficient is exactly attained by a round sphere and thus is optimal. Its proof is valid after replacing the source note's summary of coarea and genericity by an incidence-manifold argument.

Proof ledger: what was actually checked

Claim	Status	Reason / repair
Almost-everywhere regularity	VERIFIED	Define $M = \{(p,u) : N(p)\cdot u = 0\}$, a smooth compact 3-manifold; apply Sard to $M \to \mathbb{S}^2$.
Exact average spatial rim formula	VERIFIED	Apply coarea to both incidence projections and compute their 2-Jacobians explicitly.
Ellipse perimeter estimate	VERIFIED	Jensen yields $P \le \pi\sqrt{2(\kappa_1^2+\kappa_2^2)}$ with the required coefficient.
Energy identity	VERIFIED	Use $\kappa_1^2+\kappa_2^2 = 4H^2 - 2K$ and $\int K = 2\pi\chi$.
Optimality	VERIFIED	Sphere radius r: $\mathcal{L} = 2\pi r$, $A = 4\pi r^2$, $\mathcal{W} = 4\pi$, $\chi = 2$.
Equality characterization	VERIFIED	Jensen equality forces equal squared principal curvatures; Cauchy–Schwarz forces constant $\lvert B\rvert$. Nontrivial compact minimal components cannot occur.
Projected contour extension	VERIFIED	The new coarea weight is $\lvert\langle Bu, u\rangle\rvert$, the absolute normal curvature, not $\lvert Bu\rvert$.

Editorial and mathematical flags

1. **F1 — Incorrect submersion wording (repaired).** The source calls $p \mapsto N(p) \cdot u$ a submersion; a real-valued smooth function on a compact surface always has critical points. Only the zero level needs to be regular for almost all views.
2. **F2 — Unjustified transversality/stratification claim (repaired).** The source appeals to a codimension-one set without constructing the relevant map or proving the stratification. Sard's theorem on the incidence manifold is sufficient.
3. **F3 — Missing coarea Jacobian calculation (repaired).** On M the relevant Jacobians are $J_2\pi_p = (1 + |Bu|^2)^{-1/2}$ and $J_2\pi_u = |Bu|(1 + |Bu|^2)^{-1/2}$.
4. **F4 — Degenerate ellipse ambiguity (repaired).** The ellipse perimeter is interpreted as the length of its parameterized image, counting a collapsed segment twice.
5. **F5 — Visibility terminology (qualified).** Spatial rim length, projected rim length with multiplicity, and visible image silhouette length are not equal in general. The corrected paper explicitly distinguishes them.
6. **F6 — Literature and priority (open).** Existing work defines and studies contour generators and apparent contours. A targeted search did not verify a previous statement of these exact averaged formulas, but absence of a search hit is not evidence of novelty.
7. **F7 — Added results and author attribution (open).** Equality classification, lower comparison, and projected identity were developed during this review. They should not be silently presented as content of the original note or submitted without authorial review.
8. **F8 — Hypothesis scope (qualified).** The proof covers compact smooth oriented surfaces without boundary in $\mathbb{R}^3$. A surface with boundary or an unbounded surface would require separate hypotheses or boundary/integrability provisions.

Independent numerical falsification attempts

For a ring torus (R, r), $R > r$, we computed both (i) the principal-curvature integrals and (ii) the lengths of directly traced rim curves for generic viewing directions followed by a directional average. Fixed-grid quadrature initially struggled near horizontal views; adaptive quadrature resolves the narrow peaks. The new projected-length formula was tested by the same two independent methods.

Torus (R, r)	$\mathcal{L}$	$\mathcal{L}_{\mathrm{pr}}$	bound $\sqrt{A\mathcal{W}/2}$
$(2, 1)$	28.23365907	23.19516093	29.99711051
$(\sqrt{2}, 1)$	21.60993419	18.17206067	23.47400755
$(3, 0.5)$	38.46296443	30.57402222	42.16912718

The independently integrated direct rim values differed from the first column by less than 2×10^{-6}; projected values differed by less than 10^{-5} at the chosen resolution. All are below the proposed upper bound. Sphere tests at radii 0.5, 1, and 3 give exact equality

analytically. See the accompanying `numerical_verification.py` and its printed results. These tests are supplementary: a finite selection of surfaces cannot establish a theorem.

Possible strengthening and practical interpretation

The full rim length obeys the two-sided bound

$$\frac{1}{\pi}\int_\Sigma |B|\,dA \le \mathcal{L}(\Sigma) \le \frac{1}{2\sqrt{2}}\int_\Sigma |B|\,dA.$$

For projected *parameterized* contours, the independent identity is

$$\mathcal{L}_{\mathrm{pr}}(\Sigma) = \frac{1}{4\pi}\int_\Sigma \int_0^{2\pi} |\kappa_1 \cos^2\theta + \kappa_2 \sin^2\theta|\,d\theta\,dA.$$

It yields $\frac{1}{2}\int |H|\,dA \le \mathcal{L}_{\mathrm{pr}} \le \mathcal{L}$; for a convex ovaloid the left inequality becomes an equality. These observations are useful when the motivating application concerns two-dimensional projected contours rather than spatial rim curves.

What remains unverified

Novelty: no exhaustive review of the classical integral-geometry or computer-vision literature was performed; no claim of original priority is endorsed. **Publication:** the manuscript has not been externally peer reviewed. **Formal verification:** the proofs were audited mathematically and with reproducible numerical examples, but not translated into a proof assistant such as Lean or Coq. **Applications:** no image-processing benchmarks or visibility-aware formulas have been established.

Sources used for background checks

- H. Federer, *Geometric Measure Theory*, Springer, 1969 (coarea theorem).
- M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice–Hall, 1976.
- S. Plantinga and G. Vegter, "Computing Contour Generators of Evolving Implicit Surfaces," *ACM Transactions on Graphics* **25** (2006), 1243–1280; publication page.

Audit disposition: approve the corrected mathematical statements under the explicit assumptions; request originality and authorship review before journal submission.